



Nachiketa Patil

MS by Research

Centre of Machine Intelligence & Data Science

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Examination	University	Institute	Year	CPI/%
MS by Research	IIT Bombay	IIT Bombay	2024-Present	7.54
B.E. [CSE]	University of Mumbai	Atharva College of Engg.	2019-23	8.86
Higher Secondary	Maharashtra State Board	Fr. Agnel Jr. College	2019	78.46%

THESIS, PROJECT & SEMINAR

• Efficient Data Selection in Collaborative Machine Learning

(Thesis Project | Advisor: **Prof. Arjun Bhagoji**)

(Jan'25–Present)

Current work:

- Developed methods to identify useful data from multiple data owners in collaborative ML settings.
- Designed scoring mechanisms to estimate the contribution of data to downstream model performance before training.
- Analyzed how variations across data sources affect model stability and performance.
- Designed methods applicable to both labeled and unlabeled data settings.

• Joint vs. Two-Stage Optimization for Temporal Sequences

(Research Project | Advisor: **Prof. Arjun Bhagoji, Prof. Francesco Bronzino(ENS Lyon)**)

(Nov'25–Present)

- Generated synthetic network traffic data by modeling temporal event sequences using a state space model.
- Extended the NetSSM backbone to jointly learn events and their timestamps instead of using a two-stage approach.
- Compared joint and two-stage training strategies and showed improved temporal representation with reduced training cost.

COURSE PROJECTS

• Adversarial Robustness via Representation Similarity Regularization, (DS 603: Robust Machine Learning)

(Guide: **Prof. Arjun Bhagoji**)

(July'25–Nov'25)

- Proposed a defense strategy replacing output-level constraints (e.g., TRADES) with representation-level similarity regularization.
- Implemented geometric (CKA, HSIC) and neighborhood-based (CKNNA, MkNN) metrics to stabilize feature manifolds.
- Demonstrated improved robustness on WideResNet-28-10 against PGD/APGD adversarial attacks.

• Unified Multi-Task Image Denoising and Deblurring, (GMR 650: Advanced Deep Learning for Image Analysis)

(Guide: **Prof. Biplab Banerjee**)

(July'25–Nov'25)

- Developed a multi-task restoration model using a frozen DINOv3 ViT backbone with LoRA adapters, enabling unified denoising and deblurring.
- Identified negative transfer in specialist models and showed that task-conditioned multi-task learning improves generalization on OOD domains (e.g., faces).
- Implemented full training pipeline (pre-training → fine-tuning → inference) and evaluated with PSNR/SSIM on SIDD, CBSD68, RealBlur, and Helen.

• Robust 2D-DPO, (CS 769: Optimization for Machine Learning)

(Guide: **Prof. Ganesh Ramakrishnan**)

(Jan'25–May'25)

- Implemented 2D-DPO, a 2-dimensional preference optimization technique for LLM alignment.
- Proposed a novel robust formulation designed to resist noisy signal degradation.
- Supported the claims using theoretical analysis and empirical results after testing with the Pythia 6.9B model.

• Emotional Analysis for Indic Languages, (CS 626: Natural Language Processing)

(Guide: **Prof. Pushpak Bhattacharya**)

(Aug'24–Dec'24)

- Trained and fine-tuned transformer-based models for multi-lingual emotion classification.
- Performed fine-grained analysis (5 emotion classes) on the BHAAV dataset.

- Benchmarked BERT, CNN and LSTM models, achieving performance comparable to modern commercial NLP systems.
- **CLIP Implementation from Scratch**, (CS 725: Foundations of Machine Learning) (Aug'24–Dec'24)
(Guide: Prof. Preethi Jyothi)
 - Implemented CLIP (ViT + Transformer) end-to-end in PyTorch.
 - Trained on Flickr30k for text–image and image–text retrieval.
 - Learned cross-modal representations using contrastive alignment losses.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, HTML, CSS, Bash
- **Libraries:** Pandas, Numpy, PyTorch, Sklearn, SciPy
- **Tools:** Git, $\LaTeX$, Hugging Face, CUDA, Weights & Biases, VS Code, HPC Cluster

WORK EXPERIENCE & INTERNSHIPS

- **Department System Administrator**, C-MInDS, IIT Bombay (Oct'24–Present)
 - Set up and maintained departmental mail server, LDAP, internal services portal, and multiple small services (e.g., room booking, helpdesk).
 - Configured GPU/CPU servers for research use, assisting in procurement, installation, and optimization of compute resources.
 - Implemented automation and optimized resource utilization workflows for servers and internal services.
 - Set up the departmental SLURM cluster for fair resource allocation and developed a web interface for seamless interaction with the system.
 - Managed physical infrastructure, including network racks, access points, UPS, and related offline systems.
- **Summer Intern**, Jawaharlal Nehru Port Trust (JNPT) (Summer 2021)
 - Developed scripts to automate daily data backup and transfer, analyzed the data, and prepared reports for operational insights.
 - Provided remote technical support to port employees for IT-related issues.
 - Performed basic maintenance of computers and printers.

POR & EXTRA-CURRICULARS

- **Department Technical Secretary**, C-MInDS, IIT Bombay (May'25–Present)
 - Assist the department with technical support and maintain the departmental helpdesk.
 - Manage mailing lists and internal portals to ensure smooth communication.
 - Conduct technical orientation sessions for freshers and support other student events.
- **Organizer**, Techconnect, C-MInDS, IIT Bombay (Dec'24 & Dec'25)
 - Led the planning and execution of poster presentations, ensuring smooth coordination and successful event delivery.
 - Collaborated with external speakers, professors, and industry professionals to deliver high-quality content to participants.
 - Managed logistics, including venue setup, catering, and communication with all event participants.
- **Social Media Head, NSS Volunteer**, Undergraduate College (2019–2022)
 - Led live streaming and managed social media content for NSS events, ensuring real-time engagement with a broad audience.
 - Coordinated content creation and engagement on platforms like Instagram and Facebook.